Oxygen diffuses through the wall of drug containers and oxidizes many drugs rendering them inactive. It is the oxygen diffusion/reaction scenario that limits the shelf-life of many pharmaceutical products. To limit the oxidation of drugs, Oxygen scavengers, like sodium bisulfite ($NaHSO_3$) are often added to the drugs. The reaction to remove oxygen is: $2HSO_3^- + O_2 \rightarrow SO_4^{2-} + H_2O$

Consider a liquid drug stored in a cylindrical polyethylene container. The container is 15 cm high and has an inner radius of 6 cm. The initial concentration of $NaHSO_3$ in the drug formulation is 1 g/lit. Neglect diffusion through the top and bottom of the container and assume that the $NaHSO_3$ reacts instantly with the O_2. The O_2 concentration in air is also always 21%; and this may be approximated as the concentration at the outer end of the container. The effective diffusivity of O_2 through polyethylene is 9×10^{-13} m^2/s and you may assume P = 1 atm, T = 20 °C, and the process operates at steady-state. Evaluate the concentration profile of O_2 in the container material. How thick must the walls of the container be to ensure that 90% of the $NaHSO_3$ remains after 1 year?

5+4=9

Steady state diff^n. O_2 conc in drug = 0

O_2 conc. in air = 21%.

Mol. wt. of SB = 104.

a) Molar ~~flow rate~~ amount of SB in the container

$= \frac{\pi D^2}{4} \times L \times C_{SB}$

$= \left(\frac{\pi \times (\overset{0.12}{\cancel{0.06}})^2}{4} \times 15\times10^{-2}\right) m^3 \times 1\,\frac{g}{Lit} \times \frac{1000\,lit}{m^3} \times \frac{1}{1000}$

$= \cancel{1.6956}$ $\cancel{0.4241}$ $\cancel{1.6956\times10^{-3}}$ $\cancel{4.078\times10^{-3}}$ $1.6956\,g$ 1.63×10^{-2} gmoles.

$\boxed{=}$

10% SB is consumed in 1 yr.

$\therefore O_2$ diffusion rate $= \frac{0.1\times \overset{1.63\times10^{-2}}{\cancel{4.078\times10^{-3}}}}{2} \frac{moles}{yr} = 8.152\times10^{-4}\,\frac{moles}{yr}$

$\boxed{m_{O_2} = \overset{2.585\times10^{-11}}{\cancel{6.465\times10^{-12}}}\,\frac{moles}{s}.}$

Diff^n of O_2 through the walls of the solid.

Gov. eq^n $\frac{d}{dr}\left(r\frac{dC_A}{dr}\right) = 0.$

$\frac{dC_A}{dr} = \frac{C_1}{r}$

$C_A = C_1 \ln r + C_2.$

BC 1 $\quad C_2 = -C_1 \ln r_i$

BC 2 $\quad C_{A0} = C_1 \ln r_0 + C_2$

B.C.s.

At $r = r_i$ $\quad C_A = 0$ (instant reactn)

At $r = r_0$, $C_A = \frac{P_A}{RT} = C_{A0}$

$= \frac{0.21\times1\,atm}{8.205\times10^{-5}\,\frac{m^3\,atm}{K.mol}\,.\,298\,K}$

$= 8.588\,\frac{mol}{m^3}.$

$$C_{A_0} = \cancel{C_A}\ C_1 \ln r_o - C_1 \ln r_i$$

$$C_1 = \frac{C_{A0}}{\ln r_o/r_i}$$

$$C_2 = -C_1 \ln r_i = -\frac{C_{A0}}{\ln r_o/r_i} \cdot \ln r_i$$

$$C_A = \frac{C_{A0}}{\ln \frac{r_o}{r_i}} \cdot \ln r - \frac{C_{A0}}{\ln \frac{r_o}{r_i}} \cdot \ln r_i$$

$$\boxed{C_A = \frac{C_{A0}}{\ln \frac{r_o}{r_i}} \cdot \ln \frac{r}{r_i}}$$

$$\dot{m}_{O_2} = -D_{AB} \left.\frac{dc_A}{dr}\right|_{r=r_i} \times (\pi D_i L)$$

$$= -D_{AB} \frac{C_1}{r_i} \pi D_i L$$

$$= -D_{AB} \cdot \frac{C_{A0}}{\ln r_o/r_i} \cdot \frac{1}{\cancel{r_i}} \times \pi \cancel{D_i} 2 L.$$

$$\overset{2.585 \times 10^{-11}}{\cancel{6.465 \times 10^{-12}}} = 9 \times 10^{-13} \frac{\times 8.588}{\ln r_o/r_i} \cdot \pi \times 2 \times 0.15$$

$$\ln \frac{r_o}{r_i} = \cancel{1.12}\ 0.2816$$

$$\frac{r_o}{r_i} = \cancel{3.085}\ 1.325$$

$$r_o = \overset{1.325}{\cancel{3.085}} r_i = \overset{1.325}{\cancel{3.085}} \times 3 \times 10^{-2}\ m.$$

$$= \cancel{0.0925\ m}\ .\ 0.03975$$

$$\therefore \text{Thickness} = r_o - r_i = \cancel{0.0625\ m = 62.5\ mm}.$$

$$= (0.03975 - 3 \times 10^{-2})\ m$$

$$= 9.757 \times 10^{-3}\ m$$

$$= 9.75\ (\approx 10)\ mm$$